Prodigy Anesthesia Quick Review

Sleep Apnea

What is the definition of hypopnea?	Hypopnea is a 50% decrease in airflow for more than 10 seconds that occurs more than 15 seconds per hour of sleep that is sufficient to result in a 4% decrease in oxygen saturation. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1053.*
What is the definition of apnea?	10 seconds or more of complete cessation of breathing. *Nagelhout JJ, Plaus KL. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 1053.*
What are the characteristics that define obstructive sleep apnea (OSA)?	Periodic complete or partial obstruction of the airway during sleep with frequent episodes of apnea or hypopnea, and snoring. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 921-922.*
What are factors that are commonly associated with an increased risk of sleep apnea?	Body mass index greater than 35, neck circumference > 15.9 in., hyperinsulinemia, elevated glycosylated hemoglobin, and anatomic abnormalities of the upper airway. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1278.*
What physical abnormalities result from OSA?	Hypoxemia, hypercapnia, pulmonary hypertension, systemic hypertension, polycythemia, right ventricular failure, and an increased risk for ischemic cardiac and cerebrovascular disease. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1278.*
A patient with sleep apnea is presenting for surgery at an ambulatory surgery center. What is the ASA recommendation for determining whether or not the surgery should be performed in an outpatient setting?	Patients with OSA can be safely managed at outpatient surgical centers, but they have an increased risk of postoperative complications. This requirement for additional monitoring, radiology and laboratory services, and ventilatory support would make office-based procedures potentially unwise. *Butterworth JF, Mackey DC, Wasnick JD. Morgan & Mikhail's Clinical Anesthesiology. 5th ed. New York, NY: McGraw-Hill; 2013: 923.*
How does the distribution between males and females compare with obstructive sleep apnea and obesity hypoventilation syndrome?	Males have a greater incidence of OSA, but the incidence of OHS is equal for males and females. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 595.*

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What is obesity hypoventilation syndrome?	This is a syndrome that results from chronic OSA and occurs in about 5-10% of patients with OSA. It is characterized by obesity, awake arterial hypercapnia, and insufficient alveolar impairment independent of any other pulmonary disease. It progresses to right ventricular failure. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 1278.*
According to the ASA guidelines, what patients with sleep apnea must be admitted to the hospital for monitoring of their respiratory status?	Children under the age of 3 who have had a tonsillectomy and patients who have undergone uvulopalatoplasty. It is also recommended for patients with sleep apnea who have significant co-morbidities. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 595.*
Studies indicate that the use of CPAP prior to anesthesia reduces the risk of postoperative sleep apnea episodes. What explains this?	The use of CPAP preoperatively has been explained to decrease the incidence of postoperative sleep apnea by providing better sleep prior to anesthesia and reducing the hypersomnolence due to sleep deprivation. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 595.*
What anesthetic drugs may need dosage adjustment in patients with obstructive sleep apnea?	Any drug with a risk of respiratory depression such as sedatives, narcotics, and inhaled anesthetics may need to be used cautiously in patients with sleep apnea. *Barash PG, Cullen BF, Stoelting RK, Cahalan MK, Stock MC, Ortega R. Clinical Anesthesia. 7th ed. Philadelphia, PA: Lippincott Williams and Wilkins; 2013: 595.*

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